Tonelli–Shanks Algorithm for Modular Square Roots

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1 Problem Statement

Given an odd prime p and an integer n, the Tonelli–Shanks algorithm computes an integer x such that:

$$x^2 \equiv n \pmod{p}$$

provided that n is a *quadratic residue* modulo p, i.e.,

$$\left(\frac{n}{p}\right) = 1$$

where $\left(\frac{n}{p}\right)$ denotes the Legendre symbol.

2 Basic Idea

The algorithm generalizes the simple case where $p \equiv 3 \pmod{4}$, in which:

$$x \equiv n^{\frac{p+1}{4}} \mod p$$

For general odd primes, we use the Tonelli–Shanks method.

3 Algorithm Steps

Let p be an odd prime and n a quadratic residue modulo p. We want to compute x such that $x^2 \equiv n \mod p$.

Step 1: Factor $p - 1$

Write:

$$p - 1 = q \cdot 2^s$$

where q is odd and $s \geq 1$.

Step 2: Find a Quadratic Non-Residue z

Find some z such that:

$$\left(\frac{z}{p}\right) = -1$$

i.e., z is not a square modulo p.

Step 3: Initialization

Define the following:

$$\begin{aligned}
c &= z^q \mod p \\
t &= n^q \mod p \\
R &= n^{\frac{q+1}{2}} \mod p \\
m &= s
\end{aligned}$$

Step 4: Iterate Until $t \equiv 1 \mod p$

While $t \not\equiv 1 \mod p$, do:

1. Find the smallest integer $i \in [1, m)$ such that:

$$t^{2^i} \equiv 1 \mod p$$

2. Set:

$$\begin{aligned}
b &= c^{2^{m-i-1}} \mod p \\
R &= R \cdot b \mod p \\
t &= t \cdot b^2 \mod p \\
c &= b^2 \mod p \\
m &= i
\end{aligned}$$

Step 5: Return R

Now $R^2 \equiv n \mod p$. The other square root is $p - R$.

4 Example: Compute $\sqrt{10} \mod 13$

Step 1: Factor $p - 1$

$$13 - 1 = 12 = 3 \cdot 2^2 \Rightarrow q = 3,\ s = 2$$

Step 2: Find a Quadratic Non-Residue

Try $z = 2$. Check:

$$\left(\frac{2}{13}\right) = 2^6 \mod 13 = 64 \mod 13 = 12 \Rightarrow -1 \Rightarrow z = 2$$

Step 3: Initialization

$$\begin{aligned}
c &= 2^3 \mod 13 = 8 \\
t &= 10^3 \mod 13 = 12 \\
R &= 10^2 \mod 13 = 100 \mod 13 = 9 \\
m &= 2
\end{aligned}$$

Step 4: Iteration

We find the smallest i such that $t^{2^i} \equiv 1 \mod 13$:

$$t = 12, \quad t^2 = 144 \equiv 1 \mod 13 \Rightarrow i = 1$$

Now update:

$$\begin{aligned}
b &= c^{2^{1-1}} = c = 8 \\
R &= 9 \cdot 8 \mod 13 = 72 \mod 13 = 7 \\
t &= 12 \cdot 64 \mod 13 = 768 \mod 13 = 1 \\
c &= 64 \mod 13 = 12 \\
m &= 1
\end{aligned}$$

Since $t \equiv 1$, we're done.

Answer

$$\sqrt{10} \mod 13 = 7 \quad \text{or} \quad 13 - 7 = 6$$

5 Complexity

- Time complexity: $O(\log p)$
- Efficient for moderate-sized primes